

Lecture 6: Evaluation Approaches in HRI

COMP3018/COMP5018: Human-Robot Interaction

Dr. Amir Aly

School of Engineering, Computing, and Mathematics

Plymouth University

Semester 2, 2025-2026



amir.aly@plymouth.ac.uk

Introduction

- Today's Topics

- 1- Evaluation approaches in HRI

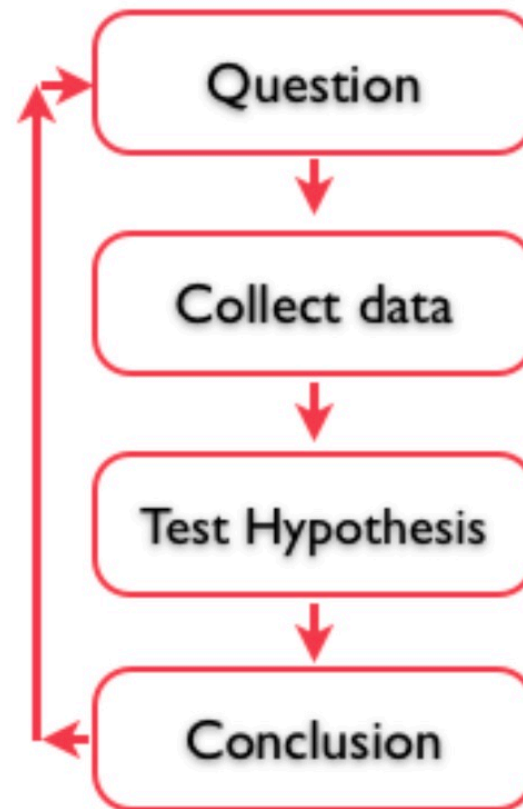
- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe the different phases of a research process

- 2- Describe the approaches evaluation in HRI subjective studies

The Research Process



1.1 The Research Problem

- Identify the research problem/question
 - Literature review
 - Open questions left unanswered by previous researchers

1.1 The Research Problem

- Defining the research problem helps you to formulate a **research hypothesis**, which is tested against the **null hypothesis**.
 - **Research hypothesis:** is the statement created by researchers when they speculate upon the outcome of a research or experiment.
 - **Null hypothesis:** a statement which the researcher tries to disprove, reject or nullify.

1.2 Data Collection

- Preparation: Make Hypothesis Testable (Operationalization)
- Preparation: Design the Study
- Conduct the Experiment or Observation

1.3 Test Hypothesis

- Organize the Data
- Analyze the Results
- Check if the Results Support your Hypothesis

1.4 Conclusion

- Look for Other Possible Explanations
- Generalize to the Real World
- Suggestions to Further Research

Evaluation Metrics

- Many possible metrics, dependent on what is being studied:
 - Percentage of tasks successfully completed
 - Accuracy
 - Time required for completion
 - Number of errors that occurred
 - Deviation from planned route/behavior/trajectory
 - Number of successfully completed subtasks
 - Fan out: number of robots controlled by a single human
 - Operator response time
 - Level of robot autonomy
 - Operator workload or effort
 - Accuracy of user mental model of the task/device
 - Level of operator trust in the system
 -

Subjective measures common in surveys and observations (Examples)

- **Likert Scale:**

Strongly Disagree	Disagree	Undecided	Agree	Strongly Agree
(1)	(2)	(3)	(4)	(5)

- **Task workload:** NASA-TLX is a subjective workload assessment tool that allows developers to perform subjective workload assessments on operator(s) working with various human-machine systems.
- **Trust:** The Working Alliance Inventory is a questionnaire commonly used in therapy and other helping relationships that tracks trust and belief in a common goal that the therapist and patient have for one another. Has been adapted to robotics as well.
- **Anxiety:** The Negative Attitudes toward Robots Scale (NARS) has been put forward as a tool for measuring people's anxiety toward robots.

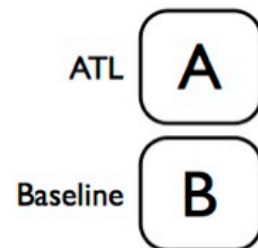
2. Performing Experimental Research

- Determine the **sampling groups** to study
 - Typically there is one **control group**, and the other groups are used to test experimental conditions
- Determine experiment design
- Conduct **pilot study**
 - small scale test of the study to ensure you're getting the kind of data you need
- Conduct the experiment
- Perform analysis

2.1 Experiment Type

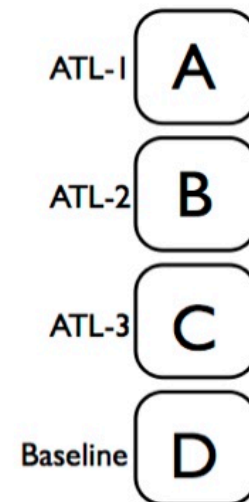
Two Group

IV: Algorithm Version



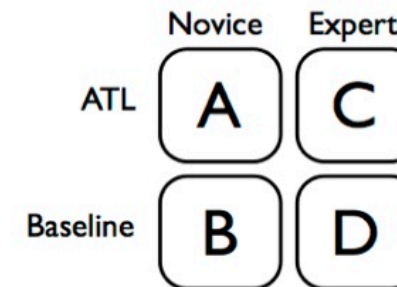
Multi-Group

IV: Algorithm Version



Factorial

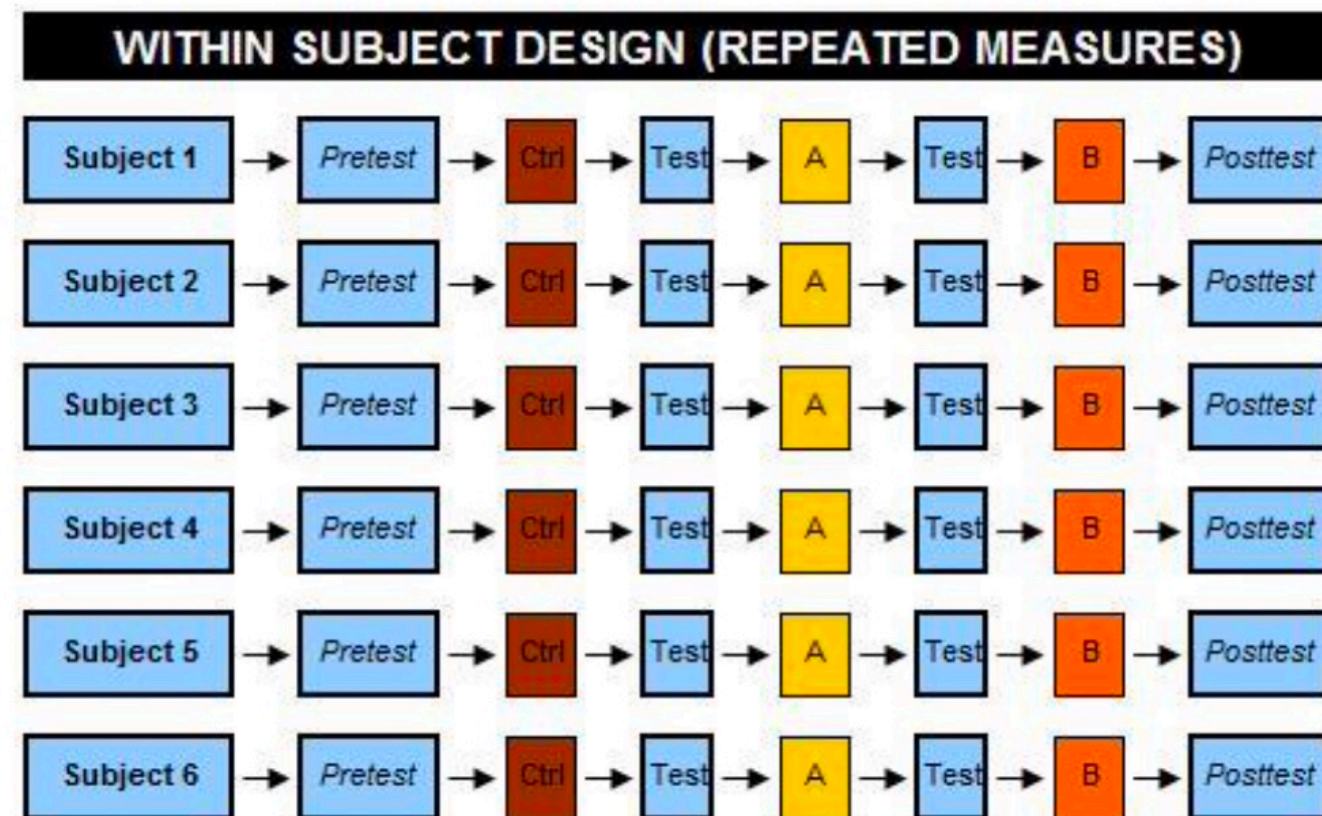
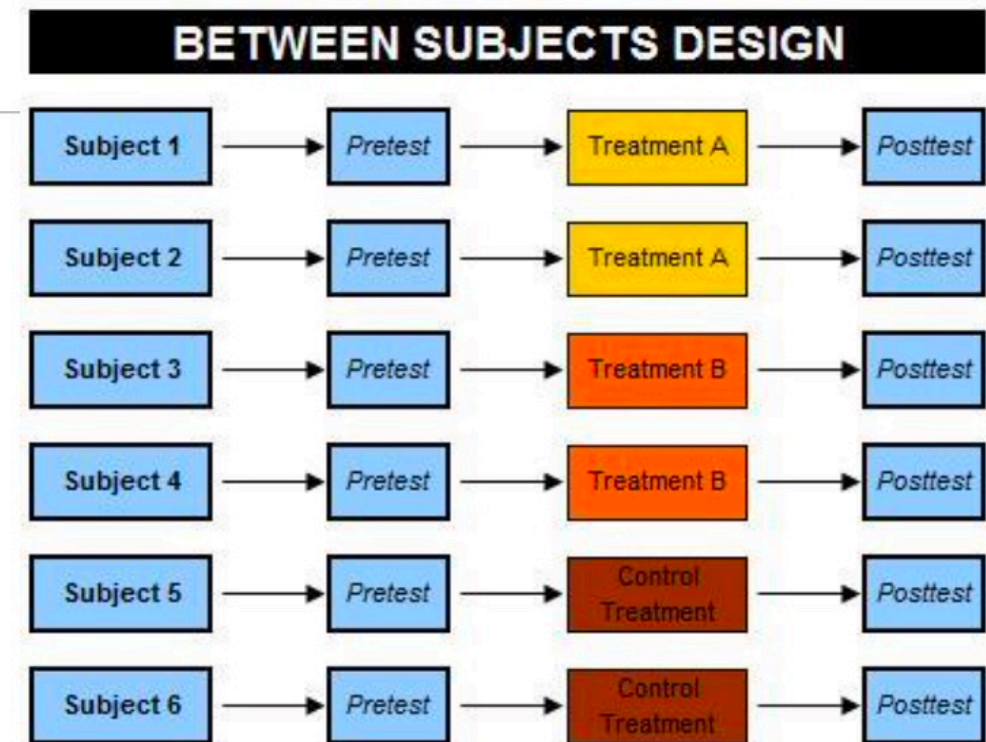
IV: Algorithm Version
Expertise



Often referred to as a 2x2 design (two independent variables, each with two possible values)

2.2 Experiment Design

- Between subjects design
 - Participants are part of a single group/treatment.
- Within subjects design
 - Every single participant is subjected to every treatment, including the control



2.2 Experiment Design (Advantages)

- Between subjects design
 - Different conditions are evaluated independently, reducing chance of one condition influencing another
 - Shorter studies, better compliance from participants
 - Easier to design
- Within subjects design
 - Fewer subjects required
 - Less chance of individuals skewing results

2.2.1 Study Variables

- The goal of good experimental design is to isolate the variables of interest such that we can draw concrete conclusions about their relationship.
 - **Independent variables** are aspects of the experiment that the experimenter will purposely manipulate in order to test these hypotheses.
 - **Dependent Variables** are aspects of the experiment that can be measured/controlled in order to see the effects of the manipulation of the independent variable(s).
- The first step in designing a study is to formulate the hypotheses in order to decide what independent and dependent variables are necessary to address them.
 - “I expect that changing [insert independent variable] will have an impact on [insert dependent variable]”

2.2.2 Novelty Effect

- Novelty effects can be particularly strong in HRI
 - Many participants may have never seen or interacted with a robot before.
 - We don't want the effects of this first interaction (so-called novelty effects) to completely overpower the manipulation of the independent variable in our study.
- One way to reduce impact:
 - Include a practice session with the robot as part of your study
 - Data from this practice session would not be recorded or used in our analysis.

2.3.1 Types of Studies: Case Studies

- an in depth study of a particular situation rather than a sweeping statistical survey.
- a study of how something works in a very specific situation
 - hard to generalize results .
 - provides in-depth analysis and useful insight for the case being examined
- Analysis: often more opinion based than statistical

2.3.2 **Types of Studies:** Longitudinal Study

- A study performed over a long period of time
 - Psychology: years and decades
 - Robotics: weeks

2.3.3 Types of Studies: Surveys

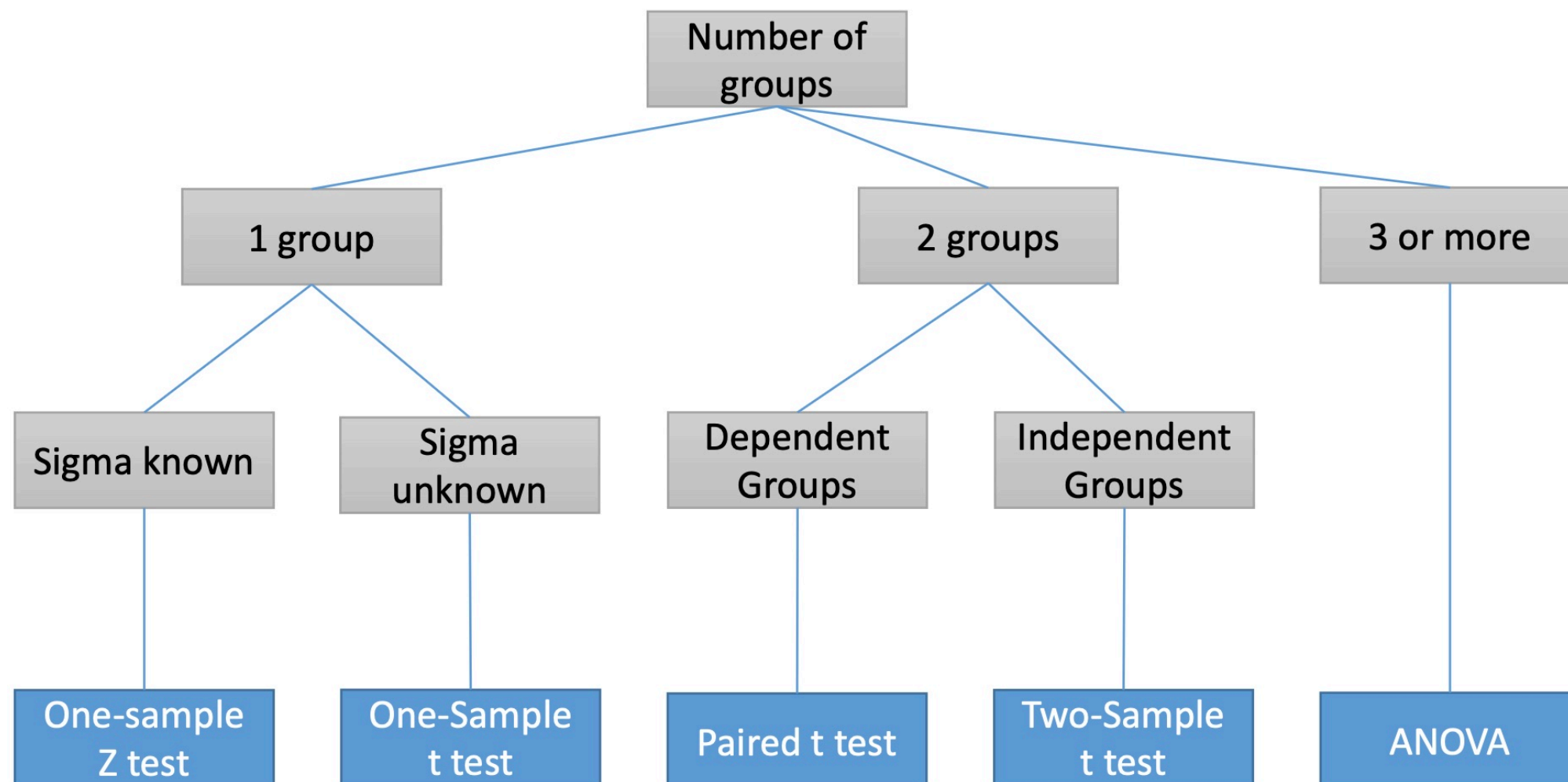
- Participants: recruited on campus, online, etc
- Types of questions: multiple choice, open answer, likert scale (“on a scale from 1 to 10...”)
- Groups of questions

2.3.4 Types of Studies: Wizard of Oz

- *laboratory studies of **simulated** systems*
- The experimenter intercepts the subject's input to the robot and may provide responses as if they were coming from the robot itself.
- Main purpose is to simulate potential future systems that
 - are expensive to prototype
 - current technologies do not yet enable

2.4 Quantitative Data Analysis

- Participants: recruited on campus, online, etc
- Types of questions: multiple choice, open answer, likert scale (“on a scale from 1 to 10...”)
- Groups of questions (to measure if there are significant differences between groups)



- T-test uses the sample's standard deviation as an estimate of the population's standard deviation

2.4 Choosing Tests for Survey Data (categorical/ordinal/continuous)

Yes/No, gender categories, survey questions, Likert

Chi-squared test

Mann-Whitney-
Wilcoxon test

t test

(Treats each category independently, thus completely ignoring the ordinal scale of Likert. Thus one of the others preferred for Likert.)

The data are interval
and normally
distributed

2.4 Choosing Tests for Survey Data (categorical/ordinal/continuous)

- **Chi-square (χ^2) test:** for **categorical counts**. Checks whether two categorical variables are **associated/independent** (i.e., whether the **distribution of counts** across categories differs between groups).
- **Mann–Whitney U (Wilcoxon rank-sum):** for **ordinal data (e.g., Likert)** or **non-normal numeric data**. Compares two independent groups using **ranks**, asking whether one group tends to have **higher values** than the other (difference in distributions / typical rank).
- **t-test:** for **continuous (interval/ratio) data** (often approximate normal, or large samples). Tests whether two independent groups have **different means** (use Welch's t-test if variances may differ).

2.4.1 P values

- P values = the probability that the observed result was obtained by chance
 - i.e. when the null hypothesis is true
- **α level** is set *a priori* (Usually 0.05)
- If **$p < \alpha$** level then we reject the null hypothesis and accept the experimental hypothesis
 - 95% certain that our experimental effect is genuine
- If however, **$p > \alpha$** level then we reject the experimental hypothesis and accept the null hypothesis

2.4.1 P values

$P < 0.05$ *

$P < 0.01$ **

$P < 0.001$ ***

Values of p	Inference
$p > 0.10$	No evidence against the null hypothesis.
$0.05 < p < 0.10$	Weak evidence against the null hypothesis
$0.01 < p < 0.05$	Moderate evidence against the null hypothesis
$0.05 < p < 0.001$	Good evidence against null hypothesis.
$0.001 < p < 0.01$	Strong evidence against the null hypothesis
$p < 0.001$	Very strong evidence against the null hypothesis

- Most authors refer to **statistically significant** as $P < 0.05$ and **statistically highly significant** as $P < 0.001$ (less than one in a thousand chance of being wrong).
- $P < 0.1$ is considered a **trend** but not statistically significant

Other Study Considerations...

Legal issues and safety

- All human subject studies require approval from the Institutional Review Board (IRB) **in the USA.**
- The IRB evaluates proposals for issues of
 - Data privacy
 - Protection of minors and vulnerable adults
 - Mental or emotional stress or humiliation
 - Physical safety
- All US government-funded projects involving human subjects require IRB review prior to funding being awarded
 - If your research is funded by NSF, NIH, etc, your project may already have completed an IRB review
 - If you are working independently, you may need an IRB review for a study

Milgram Study (Motivation for IRB)

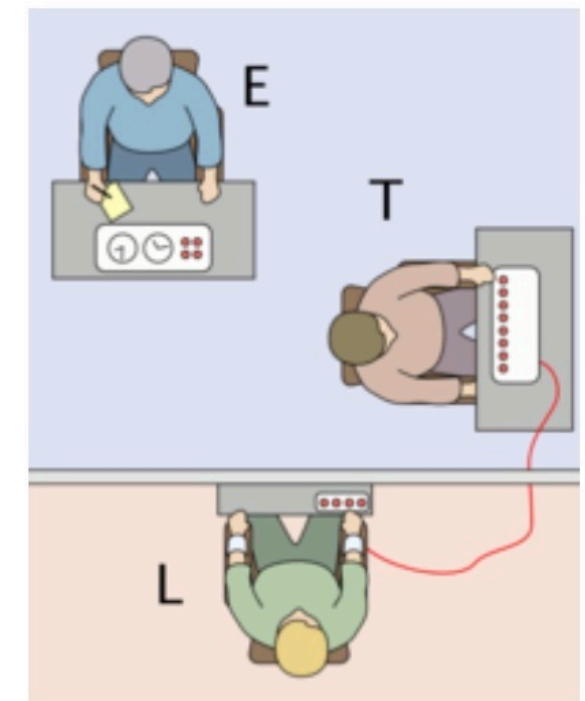
- 1960's
- Psychological study
 - series of social psychology experiments conducted by Yale University psychologist Stanley Milgram, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
- Problems
 - Psychological stress
 - Deception used – no informed consent

Public Announcement

**WE WILL PAY YOU \$4.00 FOR
ONE HOUR OF YOUR TIME**

Persons Needed for a Study of Memory

*We will pay five hundred New Haven men to help us complete a scientific study of memory and learning. The study is being done at Yale University.
*Each person who participates will be paid \$4.00 (plus 50c cashfare) for approximately 1 hour's time. We need you for only one hour; there are no further obligations. You may choose the time you would like to come (evenings, weekdays, or weekends).



Summary

- Today we discussed

- 1- Evaluation approaches in HRI

- Next week

- 1- We will discuss a different topic; i.e., Models for Human-Robot Collaboration